Mathematics Revision Test 23

There are only 10 questions from now on. This makes me Sad/Happy.
If you make a mistake please make sure you understand where you are going wrong by looking at the answers to help you with your corrections.

1. Write out the rule for finding the Area of a Triangle:

   _____ Spelling correct? Yes/No oops

2. Reduce to Lowest Terms: $\frac{9}{3}$ =_____

   24 out of 40 =_____

3. $\frac{6}{25} = \frac{\quad}{100}$

   * Remember the rule: What you do to the bottom you do to the top.

4. By how much is 8^2 greater than 3^3? _____

5. Change to Decimal Fractions:

   a $\frac{2}{50} = \frac{\quad}{100}$ =_____ b $\frac{9}{20}$ =

   x 2 (what you do to the bottom you do to the top)

6. Take 6 thousandths from 8.4 =_____

7. 78.5 ÷ 100 =_____

8. 6 x 0 x 9 x 3 =_____

9. Write 0.05 as a fraction in its Lowest Terms =_____

10. 10.04 =_____ hundredths

Are you pleased with yourself? Definitely/ Quite

DECIMAL JIGSAW

Cut out the shapes into jigsaw pieces and reassemble.
Then copy the numbers onto the blank grid.

0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09	0.1
0.11	0.12	0.13	0.14	0.15	0.16	0.17	0.18	0.19	0.2
0.21	0.22	0.23	0.24	0.25	0.26	0.27	0.28	0.29	0.3
0.31	0.32	0.33	0.34	0.35	0.36	0.37	0.38	0.39	0.4
0.41	0.42	0.43	0.44	0.45	0.46	0.47	0.48	0.49	0.5
0.51	0.52	0.53	0.54	0.55	0.56	0.57	0.58	0.59	0.6
0.61	0.62	0.63	0.64	0.65	0.66	0.67	0.68	0.69	0.7
0.71	0.72	0.73	0.74	0.75	0.76	0.77	0.78	0.79	0.8
0.81	0.82	0.83	0.84	0.85	0.86	0.87	0.88	0.89	0.9
0.91	0.92	0.93	0.94	0.95	0.96	0.97	0.98	0.99	1

DECIMAL JIGSAW

When you have completed the jigsaw, copy the numbers onto this grid. Notice the pattern of numbers.

Plotting shapes

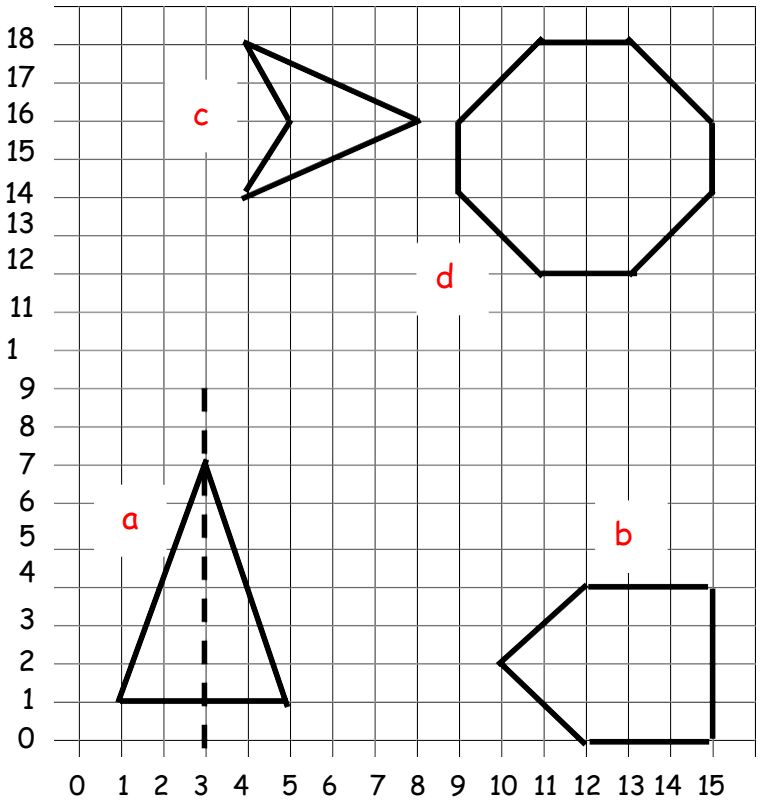
Copy the vertical and horizontal numbers in this diagram on to the squared grid on the attached sheet.

Now mark and join together, in order, the following points:

1,1 5,1 3,7 1,1

The shape drawn should be the isosceles triangle marked a in this diagram.

This shape has one axis of symmetry and it has been drawn in the shape with a dashed line.



What you have to do now….

1. Copy the other three shapes (b, c and d) onto the grid, and make a list of the coordination points plotted for each shape.

2. Mark the axes of symmetry in each shape.

3. Plot these points, joining them together in order:

   e. 1,13 4,13 4,10 1,10 1,13
   f. 6,0 9,0 9,4 6,4 6,0
   g. 12,7 12,11 15,10 15,6 12,7
   h. 5,12 6,14 8,14 9,12 8,10 6,10 5,12
   i. 5,6 11,6 9,9 5,9 5,6

Name each shape and draw in any lines of symmetry.

Use this sheet for plotting your shapes

Core Topic Timed Test 9

You have 25 minutes to complete this test.

Write the missing number in the box:

1. $$\begin{array}{r} 2\,\square\,8 \\ 3\,\overline{)\,7\;1\;4} \end{array}$$

2. $$\begin{array}{r} 9\;2 \\ 9\,\overline{)\,8\,\square\,8} \end{array}$$

3. $$\begin{array}{r} 1\;2\;3 \\ 6\,\overline{)\,7\;3\,\square} \end{array}$$

4. $$\begin{array}{r} 1\;3\;8 \\ 5\,\overline{)\,\square\;9\;0} \end{array}$$

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5. Biscuits cost £1.27 per pack. How much do 100 packs cost? _____

6. Postcards cost 49p each. How much do 1000 postcards cost? _____

7. Apples cost £1.98 a kilo. How much do 10 kilos cost? _____

Which is larger:

a. 50% of 190 or

b. $\frac{4}{5}$ of 120?

8. ____ is larger

9. The difference is _____

On a family outing Dad bought 2 adult tickets at £13.48 each and 3 tickets for the children at £8.21 each.

10. How much did he have to pay? _____

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11. There are 52 weeks in a year. The farmer recorded rain falling in 39 weeks last year. The remaining weeks had no rain. In what percentage of the weeks was no rainfall recorded?

_____

Core Topic Timed Test 9

12. $\frac{1}{4}$ = _____ % 13. $\frac{1}{2}$ = _____ % 14. $\frac{3}{4}$ = _____ %

15. $\frac{3}{5}$ = _____ % 16. $\frac{3}{1}$ = _____ % 17. $\frac{1}{3}$ = _____ %

Work out these sums:

18. 4(1.96 + 3.47) = _____ 19. 6(2.75 + 6.29) = _____

20. 3(8.09 + 7.51) = _____ 21. 9(1.81 + 5.32) = _____

Find the mean (mean means <u>average</u>) for each row:

22.	7	21	3	15				Answer: _____
23.	112	43	64	102	84			Answer: _____
24.	8	6	9	11	5	8	9	Answer: _____
25.	£1.43	93p	£1.08	79p	57p			Answer: _____

A kilo of oranges costs £1.20. If b is the number of kilos bought and a is the total price paid, write the formula for finding the total cost.

26.___

Draw a ring round the number which is 20% of 2400.

27. 244 440 1200 1400 600 280 480

The BODMAS Rule

The rule for the order in which you conduct calculations is:

BODMAS

This means:
Brackets

Orders

Division

Multiplicatio

n **A**ddition

Subtraction.

In other

words:

1. You must work out the sum in the **B**racket first and then complete any **O**rders (this also means indices, powers, or roots e.g. 3^2).

2. Next you must work out any **D**ivison or **M**ultiplication.

3. Finally complete any **A**ddition or **S**ubtraction required.

Remember - if a number appears immediately before a bracket it means you must **M**ultiply (the result of the sum in the bracket) by that figure, so:
5(3.1 + 1.9) means 5 x (3.1 + 1.9) = 5 x 5 = 25

Here is an example of the **BODMAS** rule in action:

6(1.2 + 0.8) – (4.6 - 1.6) Do the **B**racket sums first
= 6 x 2 – 3 Then **M**ultiply
= 12 – 3 Then **S**ubtract
= 9 ✓

Now look at this question:

6(2.36 + 5.62) =

6(2.36 + 5.62) = Do the **B**racket sum first - 2.36 + 5.62 = 7.98
6 x 7.98 = 47.88 ✓ Then **M**ultiply

ANSWERS - COMPLETE MATHS – PART 24

Maths

Revision Sheet

1. Half the base x the height
2. $\frac{3}{10}$ $\frac{3}{5}$
3. $\frac{24}{100}$
4. 37
5. 0.54 0.45
6. 8.394
7. 0.785
8. 0
9. $\frac{1}{20}$
10. 1004

Question 5:

Changing fractions to decimals:
You have to put the bottom number over 100 first by either x that number by 10 or any other number needed to achieve that aim. In this case 2 is used. Then remember the rule – What you do to the top you do to the bottom.
Therefore:

$$\frac{27}{50} = \frac{54}{100} = 0.54$$

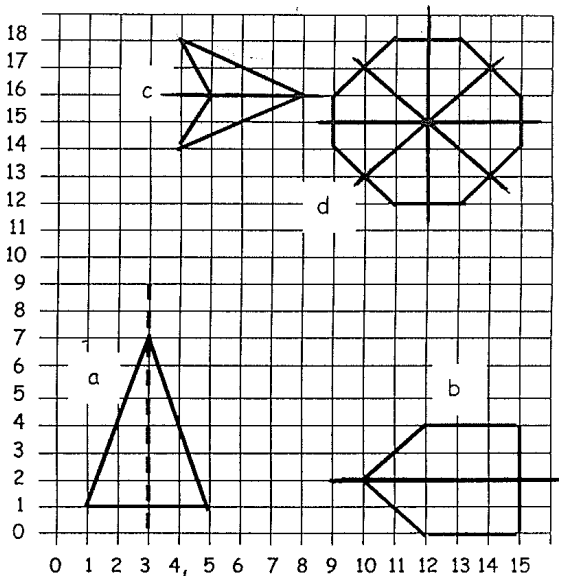
Got it?

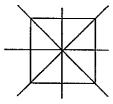
Plotting Shapes

1. Shape b: (12,0) (15,0) (15,4) (12,4) (10,2)
   Shape c: (4,14) (8,16) (4,18) (5,16)
   Shape d: (11,12) (13,12) (15,14) (15,16) (13,18) (11,18) (9,16) (9,14)

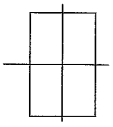
This is an irregular octagon as not all sides are the same length. So it only has 4 lines of symmetry. A regular octagon will have 4 lines of symmetry.

2. Symmetry lines are shown here:



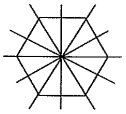
3. e. Square 

   (4 lines of symmetry)

   f. Rectangle 

   (2 lines of symmetry)

   g. Parallelogram - no lines of symmetry

   h. Hexagon 

   (6 lines of symmetry)

   i. Quadrilateral - no lines of symmetry

ANSWERS - COMPLETE MATHS – PART 24

Core Topic Timed Test 9

1. 3
2. 2
3. 8
4. 6
1. £127
2. £490
3. £19.80
4. b
5. 1
6. £51.59
7. 25%
8. 25%
9. 50%
10. 75%
11. 60%
12. 30%
13. $33\frac{1}{3}$ %
14. 21.72 (see Bodmas Rule on next sheet)
15. 54.24 (see Bodmas Rule on next sheet)
16. 46.8 (see Bodmas Rule on next sheet)
17. 64.17 (see Bodmas Rule on next sheet)
18. 11.5
19. 81
20. 8
21. 96p
22. a = £1.20 x b OR:
    a = b(£1.20) (see Bodmas Rule)
23. 480